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Principles of Macroeconomics

Lecture 11: Aggregate Demand and Aggregate Supply

Prof. Dr. Jan-Egbert Sturm

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Lecture 10: Keynesian Economics and IS-LM Analysis

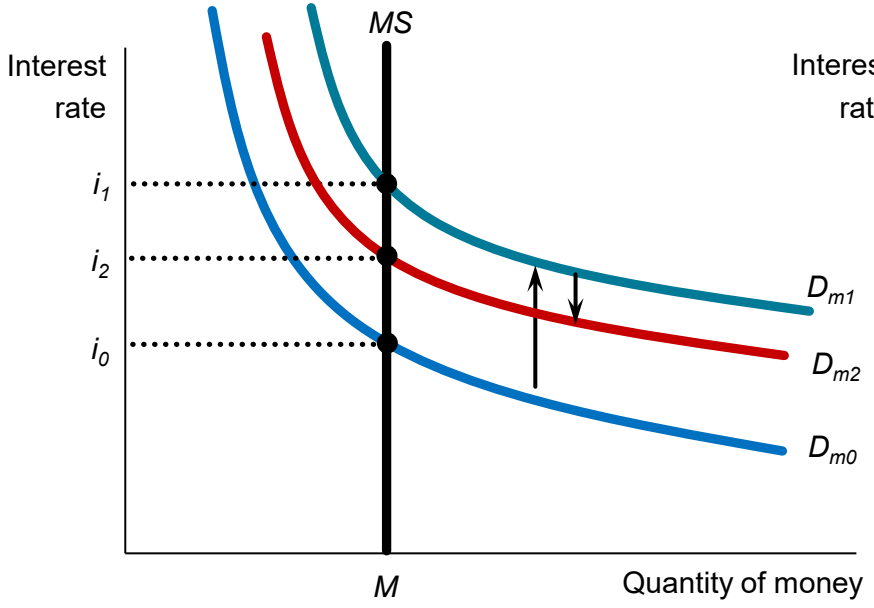
Prof. Dr. Jan-Egbert Sturm

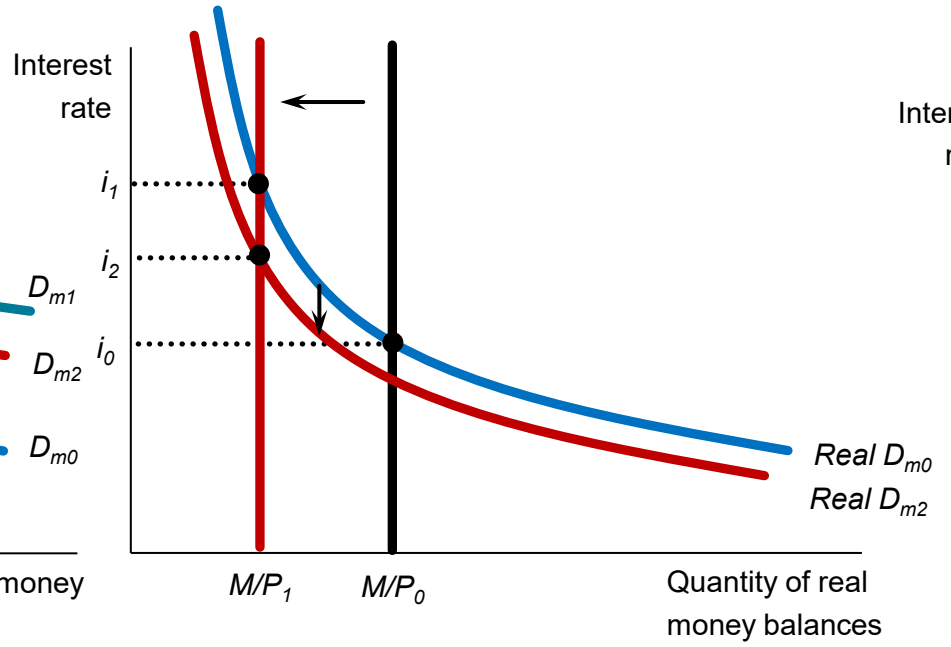
From IS-LM to Aggregate Demand

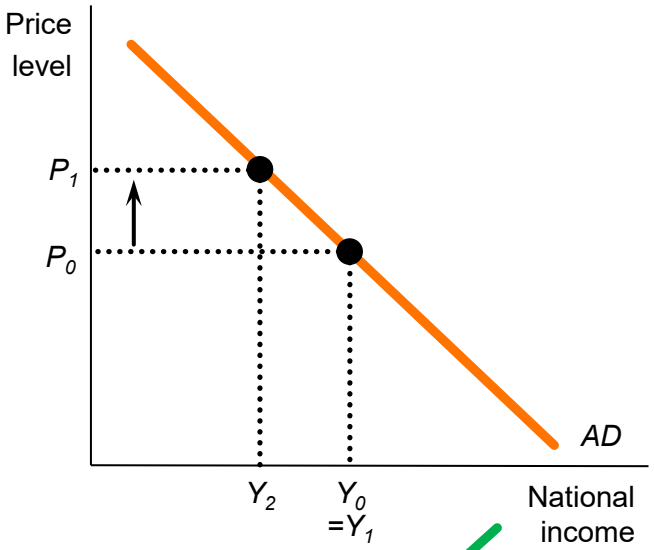
- It is a small step from the IS-LM model to the aggregate demand and aggregate supply model
  - Aggregate demand shows the relationship between national income and the price level
- Note that we have so far assumed prices to be stable/constant/sticky
  - Hence, P was fixed
  - We have therefore been referring to the real money supply – what money can actually buy
  - Real money supply (M/P) is given by M (nominal money supply) divided by P, the price level
- A rise in P causes a fall in M/P, shifting the LM curve to the left
- The result is that national income falls and interest rates rise
- Hence, higher P lowers aggregate demand

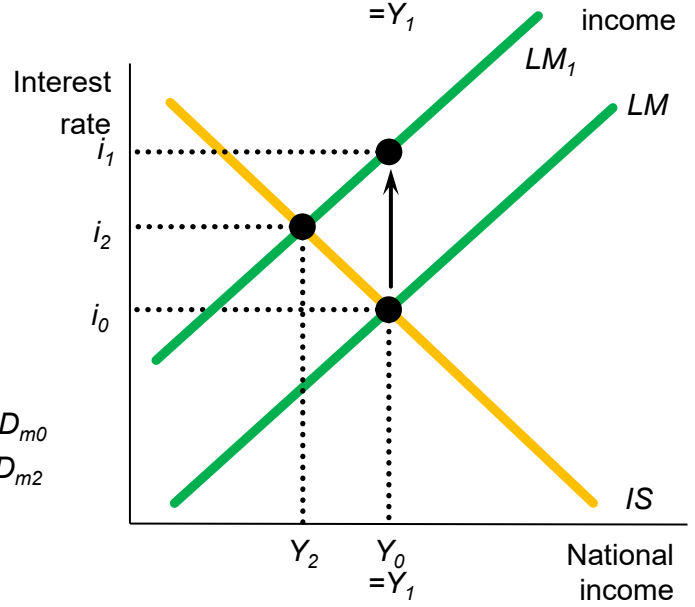
General Equilibrium

1. A rise in the price level reduces real money balances and shifts the LM curve to the left
   - This raises the equilibrium interest rate and lowers national income
2. The aggregate demand curve AD shows the inverse relationship between the price level and national income





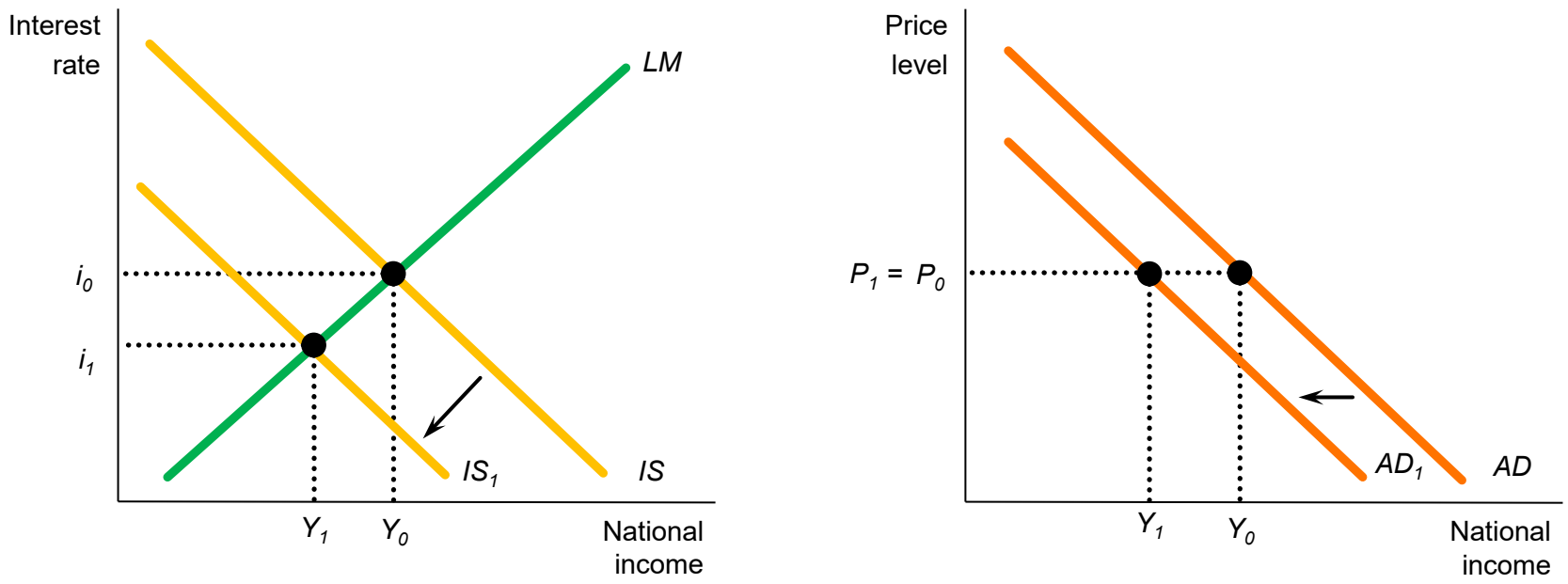




From IS-LM to Aggregate Demand

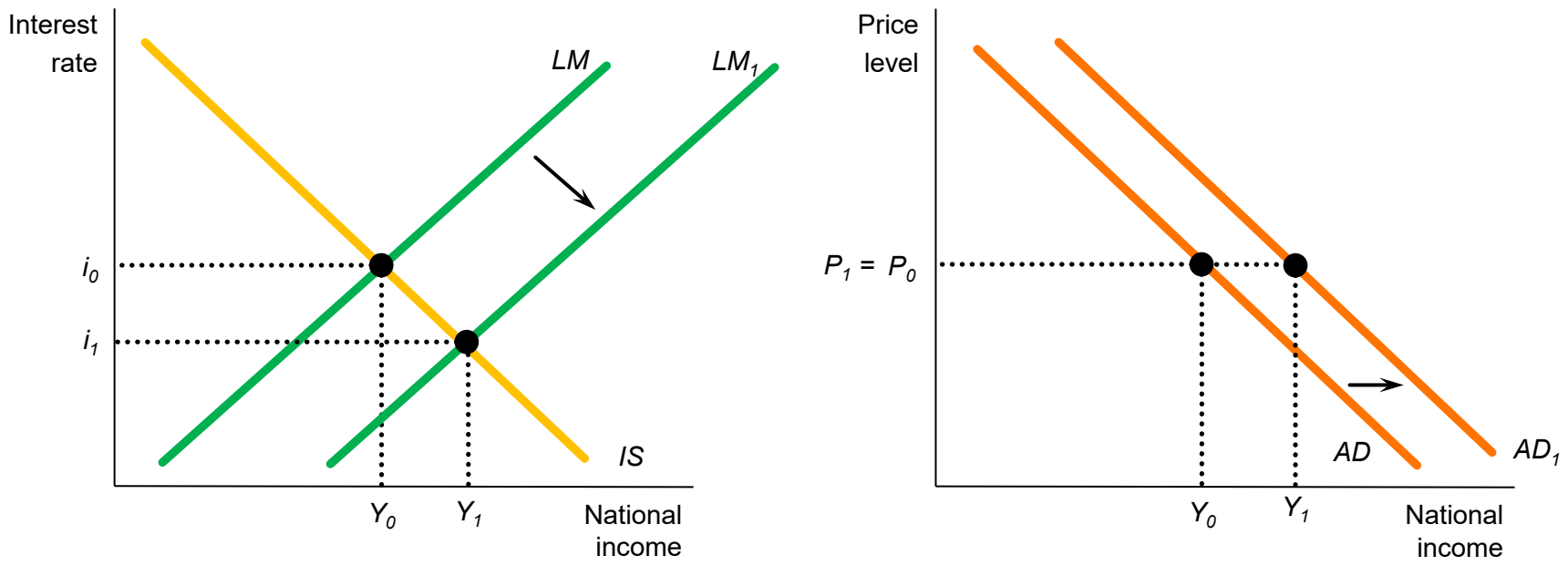
- Assume the price level remains constant, then a change in national income in the IS-LM model will result in a shift in the aggregate demand curve
- For example:
  - Austerity measure of tightening fiscal policy causes a fall in national income as aggregate demand shifts inwards
  - Central bank loosens money supply, e.g. through quantitative easing, boosts national income as aggregate demand shifts outwards

A Government Tightening Fiscal Policy



- In the short term, tightened fiscal policy reduces national income and the economy's interest rate, but the price level (is assumed to) remain(s) constant

Central Bank Loosens Monetary Policy



- In the short term, expansive monetary policy increases national income and reduces the economy's interest rate, but the price level (is assumed to) remain(s) constant

A Return to Keynesianism?

- The severity of the global recession, the euro crisis and the Corona pandemic has led to calls for fiscal stimulation especially as monetary policy has failed to stimulate growth
- Two views
  - To raise the money governments will either have to tax its citizens more or increase borrowing and therefore crowding out private investment
  - The depth of some of these economic crises was/is such that a fiscal stimulus was/is necessary to get out of a liquidity trap and, far from crowding out, fiscal stimuli would lead to ‘crowding in’

Summary

- J.M. Keynes's primary message was that recessions and depressions can occur because of inadequate aggregate demand.
- He advocated governments intervene to boost demand through influencing aggregate demand.
- The Keynesian cross diagram shows how the economy can be in equilibrium when E = Y.
- This equilibrium may not be sufficient to deliver full employment output and so the government can attempt to boost demand to help achieve full employment.
- John Hicks developed Keynes's ideas in the form of the IS-LM model which shows general equilibrium in the economy.
- The IS (investment–saving) curve shows all points of equilibrium in the goods market at a particular interest rate and level of national income.

Summary

- The LM (liquidity–money supply) curve shows points where the money market is in equilibrium at particular rates of interest and level of national income.
- General equilibrium occurs where the IS curve intersects the LM curve. At this interest rate and level of national income both the goods market and money market are in equilibrium.
- Fiscal policy and monetary policy can cause shifts in the IS and LM curves bringing about new equilibrium positions. The outcome will depend on a variety of factors, including the response of consumption and investment to changes in interest rates and the public's response to holding monetary balances as a result of a change in interest rates.
- From the IS-LM model we can derive the aggregate demand curve that represents the relationship between national income and the price level.
- The IS-LM model has been developed to incorporate change in the way policy is conducted, with more focus on inflation targeting.

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Principles of Macroeconomics

Lecture 11: Aggregate Demand and Aggregate Supply

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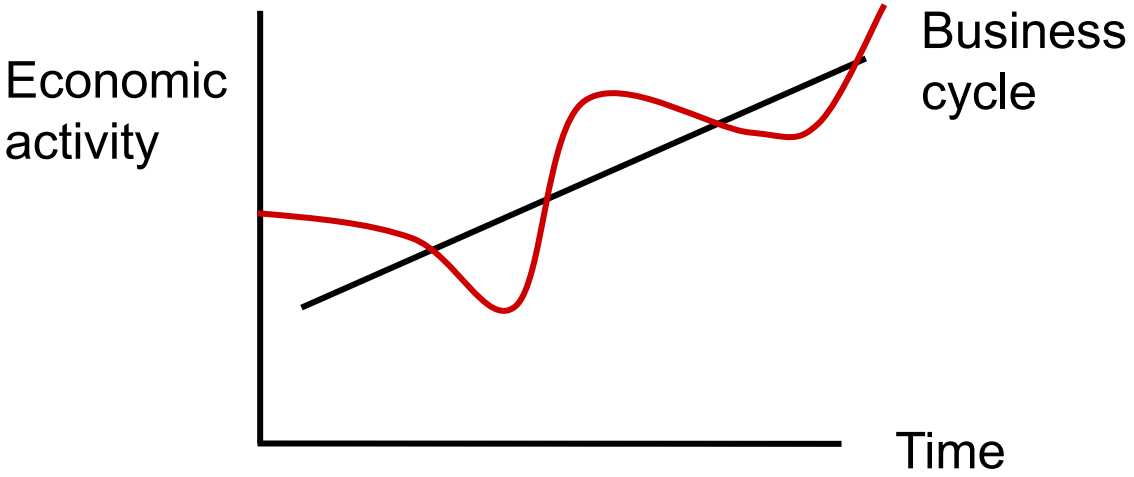
Explaining Short-Run Economic Fluctuations

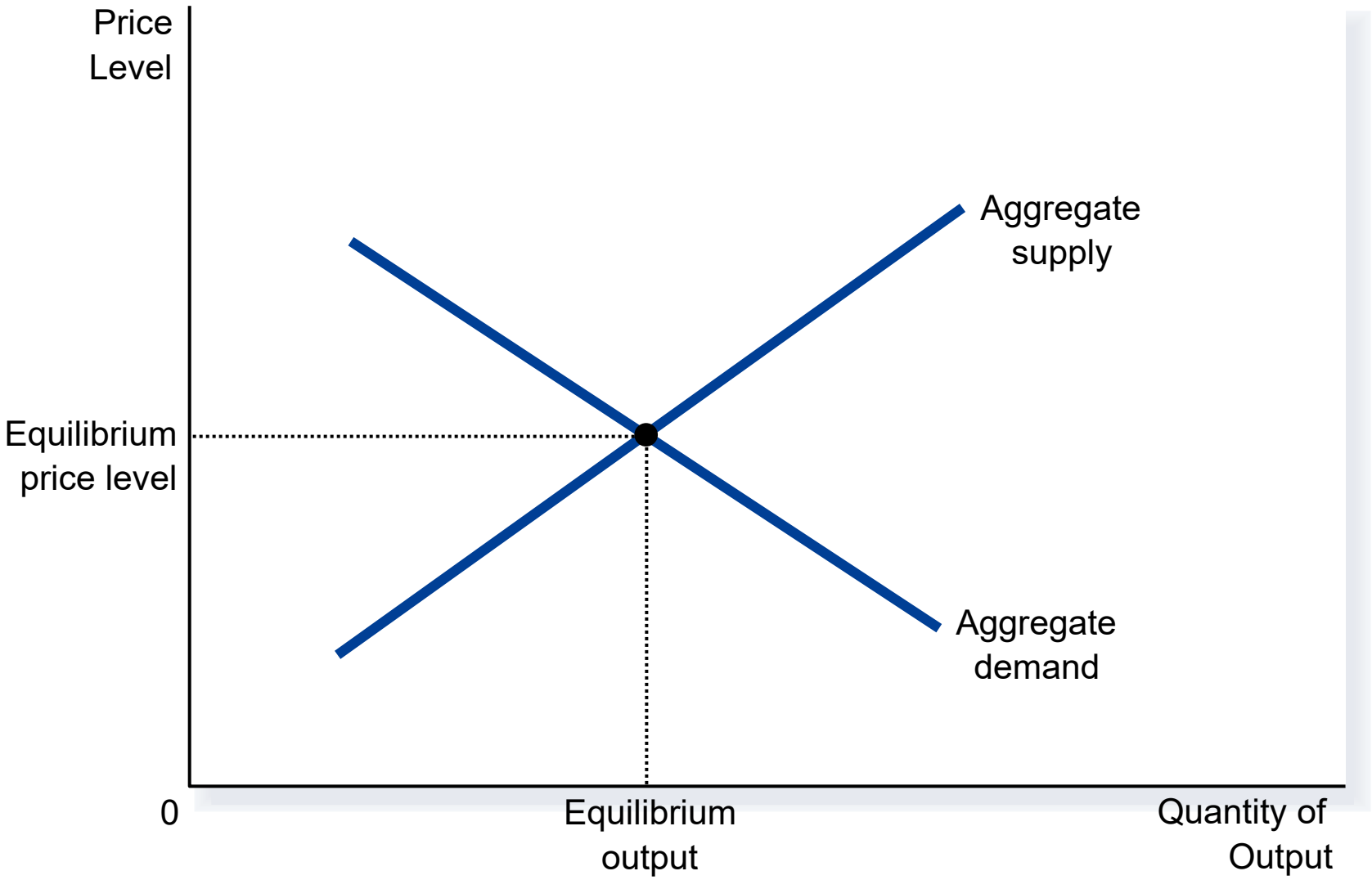
- The Assumptions of Classical Economics
  - Most economists believe that classical theory describes the world in the long run, but not in the short run
  - Changes in the money supply affect nominal variables but not real variables in the long run
  - The assumption of monetary neutrality is not appropriate when studying year-to-year changes in the economy
- Neutrality of money
  - If the quantity of money in the economy were to double, prices would double and so would (nominal) incomes
    - Real variables would remain constant
  - HOWEVER: These changes will not occur instantaneously
    - It takes time for prices and incomes to change, and in the meantime, there can be real effects

The Model of Aggregate Demand and Aggregate Supply

- Two variables are used to develop a model to analyze the short-run fluctuations
  - The economy's output of goods and services measured by real GDP
  - The average level of prices measured by the CPI or the GDP deflator
- Economist use the ***model of aggregate demand and aggregate supply*** to explain short-run fluctuations in economic activity around its long-run trend
  - The ***aggregate-demand curve*** shows the quantity of goods and services that households, firms, and the government want to buy at each price level
  - The ***aggregate-supply curve*** shows the quantity of goods and services that firms choose to produce and sell at each price level



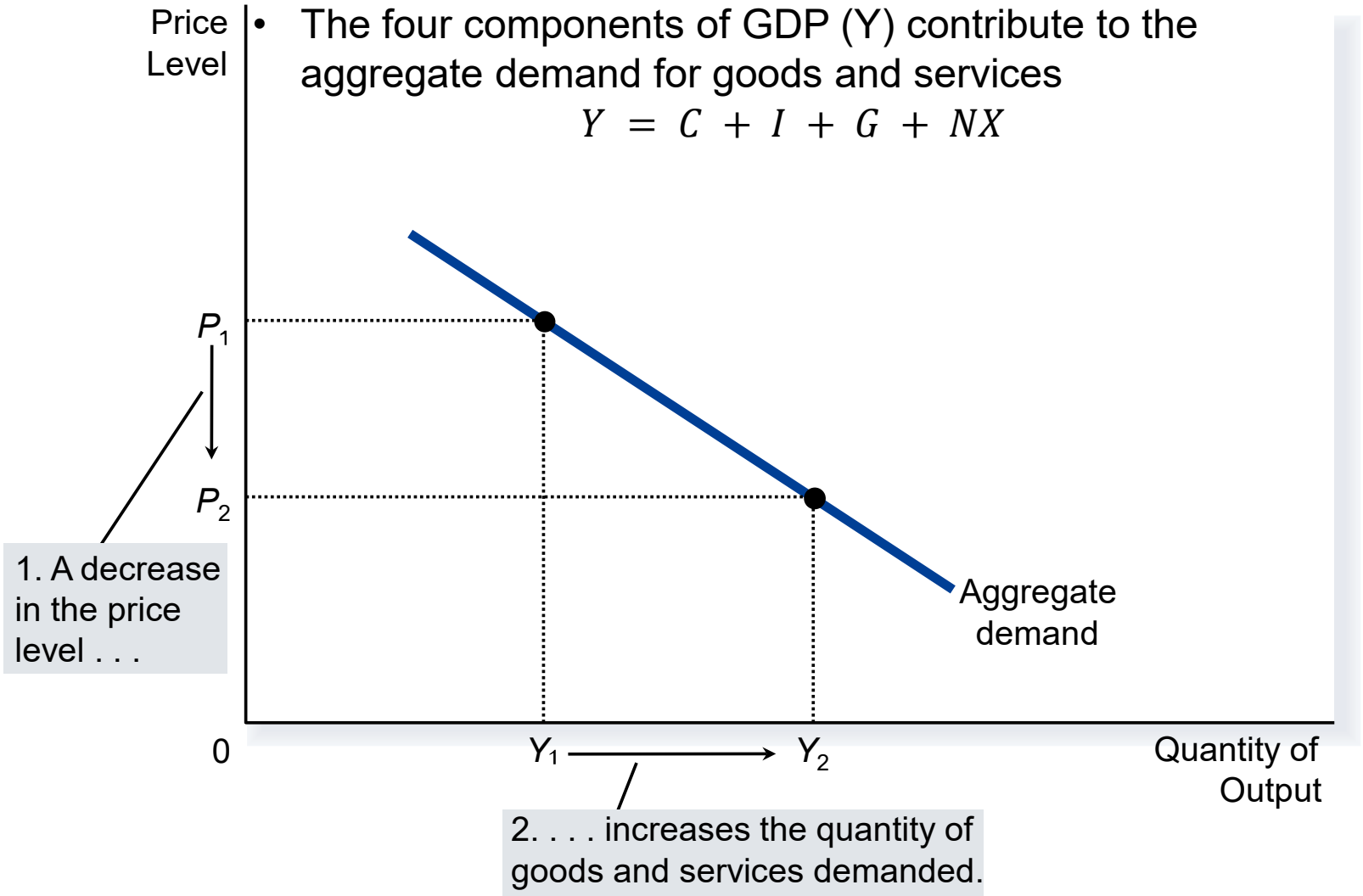
Aggregate Demand and Aggregate Supply



The Aggregate-Demand Curve...

- The four components of GDP (Y) contribute to the aggregate demand for goods and services

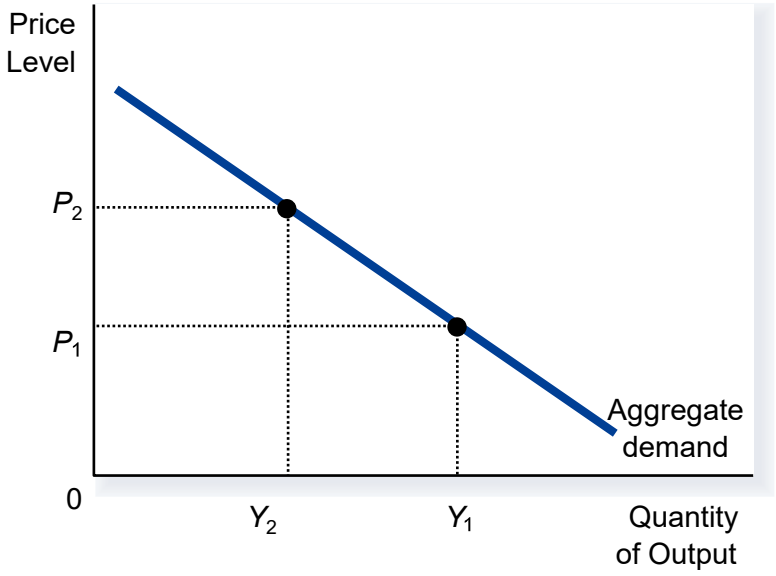
$$Y = C + I + G + NX$$



Why the Aggregate-Demand Curve Is Downward Sloping

- The Price Level and Consumption:
  - The Wealth Effect
- The Price Level and Investment:
  - The Interest Rate Effect
- The Price Level and Net Exports:
  - The Exchange-Rate Effect

The Aggregate-Demand Curve



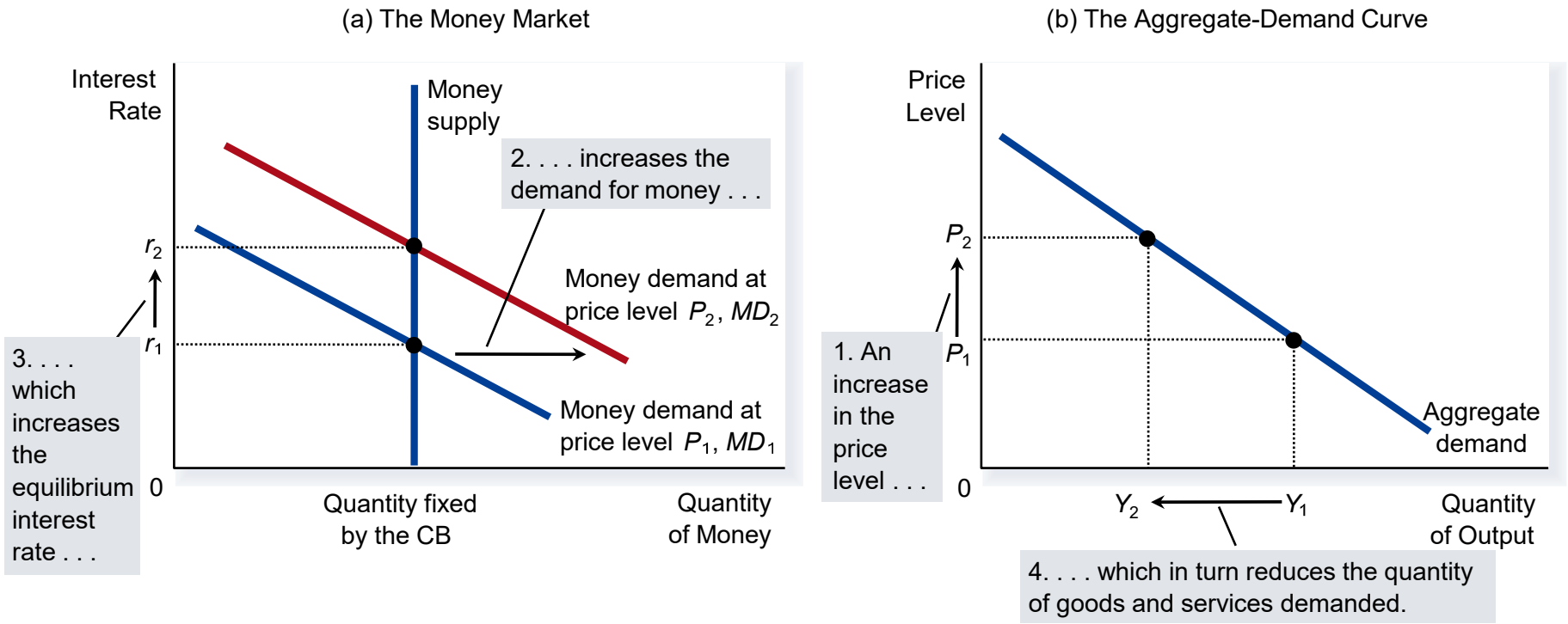
Why the Aggregate-Demand Curve Is Downward Sloping

- The Price Level and Consumption:
  - The Wealth Effect
    - An increase in the price level makes consumers feel less wealthy, which in turn encourages them to spend less
    - This decrease in consumer spending means smaller quantities of goods and services demanded

Why the Aggregate-Demand Curve Is Downward Sloping

- The Price Level and Consumption:
  - The Wealth Effect
- The Price Level and Investment:
  - A higher price level implies
    - Buying goods and services requires more money
    - At given interest rate the demand for money increases
    - Given the amount of money in circulation, the price of holding money needs to increase, i.e. the interest rate goes up

The Money Market and the Slope of the Aggregate Demand Curve (link price level – interest rate)



Why the Aggregate-Demand Curve Is Downward Sloping

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    - At given interest rate the demand for money increases
    - Given the amount of money in circulation, the price of holding money needs to increase, i.e. the interest rate goes up
  - The Interest Rate Effect
    - A higher price level increases the (real) interest rate and makes borrowing more expensive, which discourages spending on investment goods
    - This decrease in investment spending means a smaller quantity of goods and services demanded

Why the Aggregate-Demand Curve Is Downward Sloping

- The Price Level and Consumption:
  - The Wealth Effect
- The Price Level and Investment:
  - The Interest Rate Effect
- The Price Level and Net Exports:
  - The Exchange-Rate Effect
    - A higher price level in Switzerland causes (Swiss interest rates to increase and) the real exchange rate to appreciate, which depresses Swiss net exports.
    - The decrease in net export spending means a smaller quantity of goods and services demanded

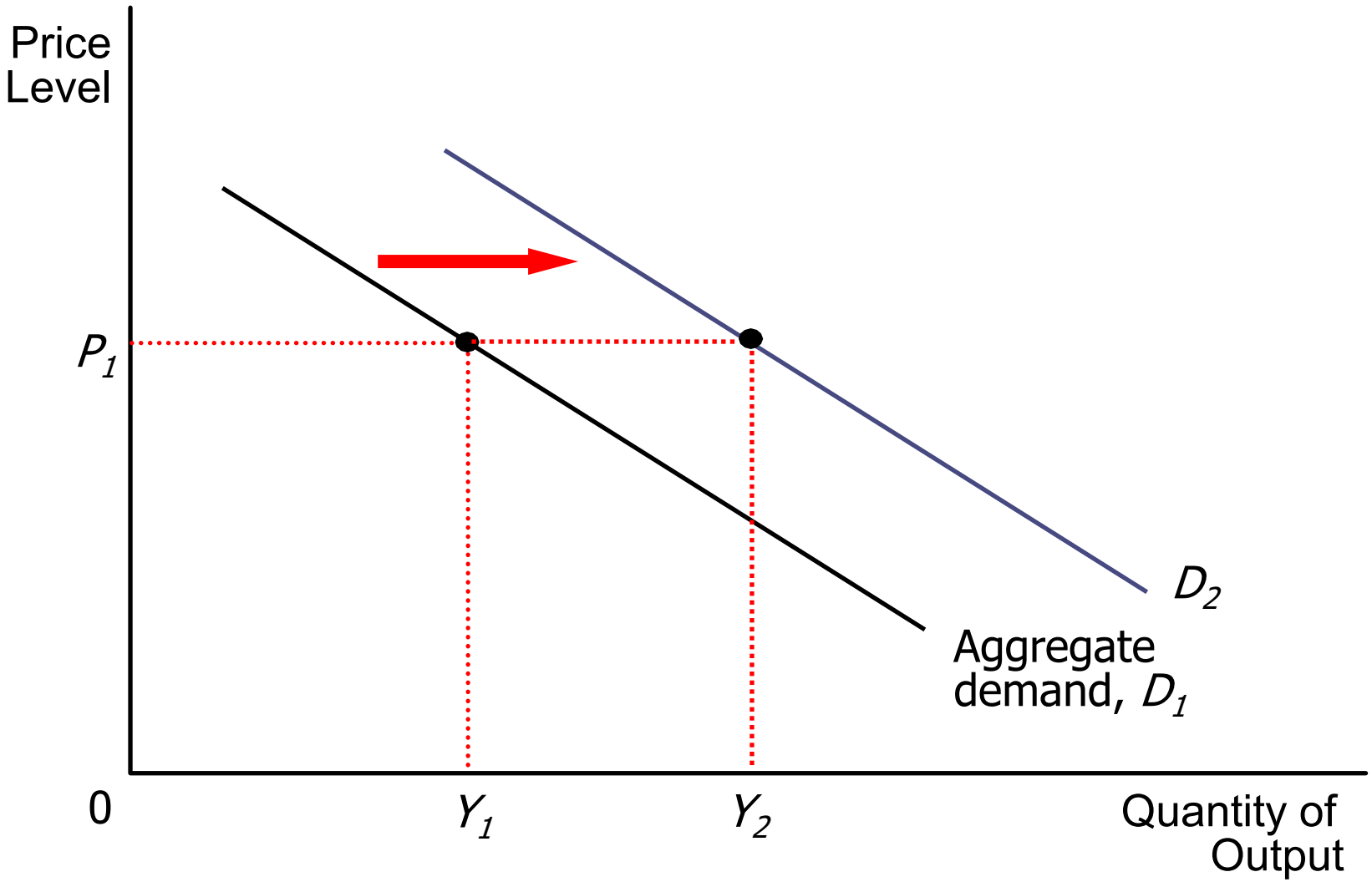
Why the Aggregate-Demand Curve Is Downward Sloping

1. The wealth effect
   - A higher price level decreases the real value of money and makes consumers less wealthy, which discourages them to spend more
   - This decrease in consumer spending means smaller quantities of goods and services demanded
2. The interest-rate effect
   - A higher price level increases the real interest rate and makes borrowing more expensive, which discourages spending on investment goods
   - This decrease in investment spending means a smaller quantity of goods and services demanded
3. The exchange-rate effect
   - A higher price level in Switzerland causes (Swiss interest rates to rise and) the real exchange rate to appreciate, which depresses Swiss net exports
   - The decrease in net export spending means a smaller quantity of goods and services demanded

Why the Aggregate-Demand Curve Might Shift

- The downward slope of the aggregate-demand curve shows that a fall in the price level raises the overall quantity of goods and services demanded
- Many other factors also affect the quantity of goods and services demanded at any given price level
- When one of these other factors changes, the aggregate demand curve shifts
- Shifts might arise from changes in:
  - (autonomous) Consumption
  - (autonomous) Investment
  - Government Purchases
  - (autonomous) Net Exports
  - Money Supply

Shifts in the Aggregate Demand Curve

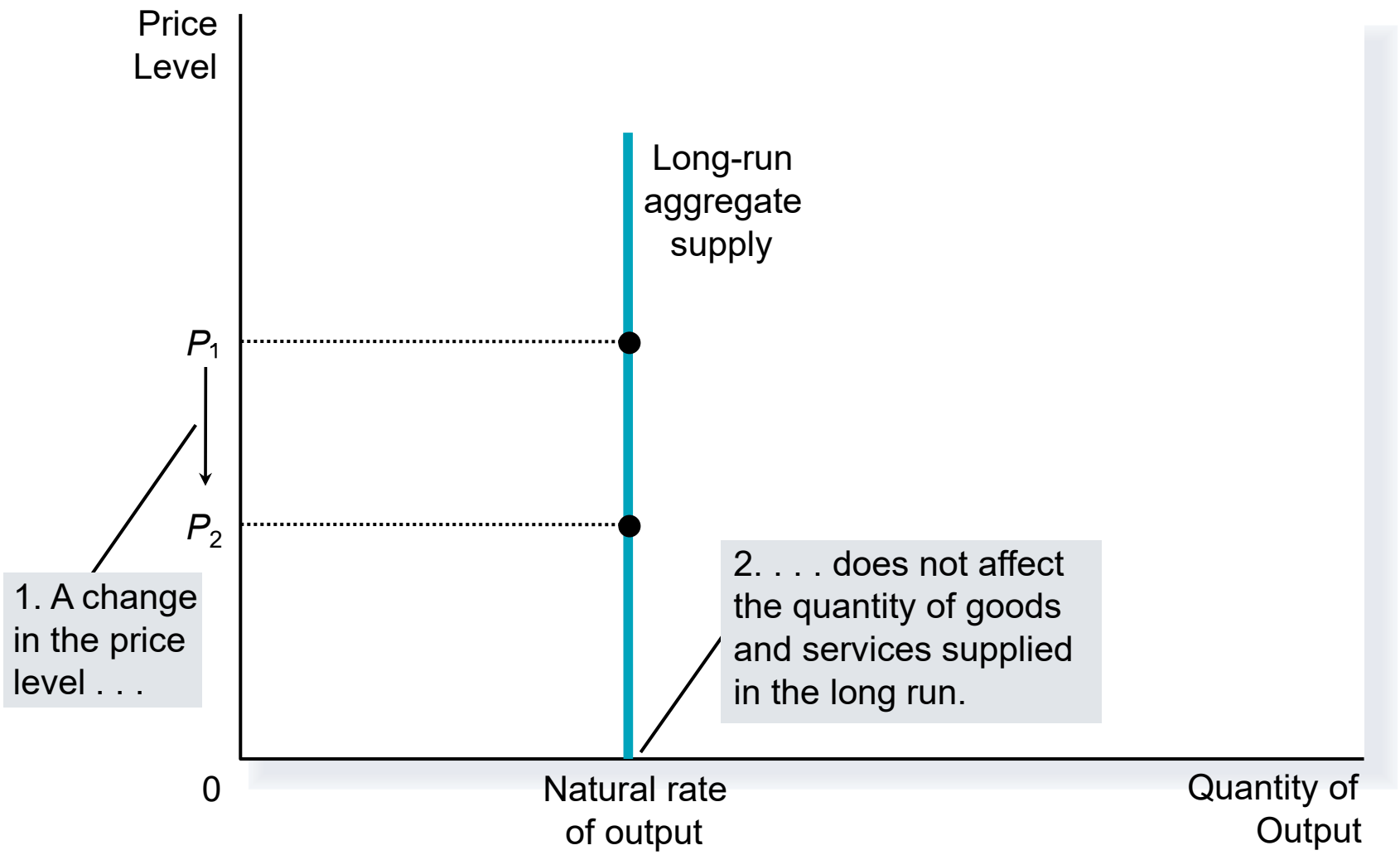


The Aggregate-Supply Curve

- In the long run, the aggregate-supply curve is vertical because the price level does not affect long run determinants of real GDP
  - In the long run, an economy's production of goods and services depends on its supplies of labour, capital, and natural resources and on the available technology used to turn these factors of production into goods and services
  - The price level does not affect these variables in the long run
  - The long-run aggregate supply represents the classical dichotomy and money neutrality
- The long-run aggregate-supply curve is vertical at the natural rate of output, which is the production of goods and services that an economy achieves in the long run when unemployment is at its normal rate
  - This level of production is also referred to as potential output or full-employment output
  - The natural rate of output is the level of output towards which the economy gravitates in the long run

- In the short run, the aggregate-supply curve is upward sloping
  - In the extreme short run (IS/LM model) it is even horizontal (P is given/fixed/constant)

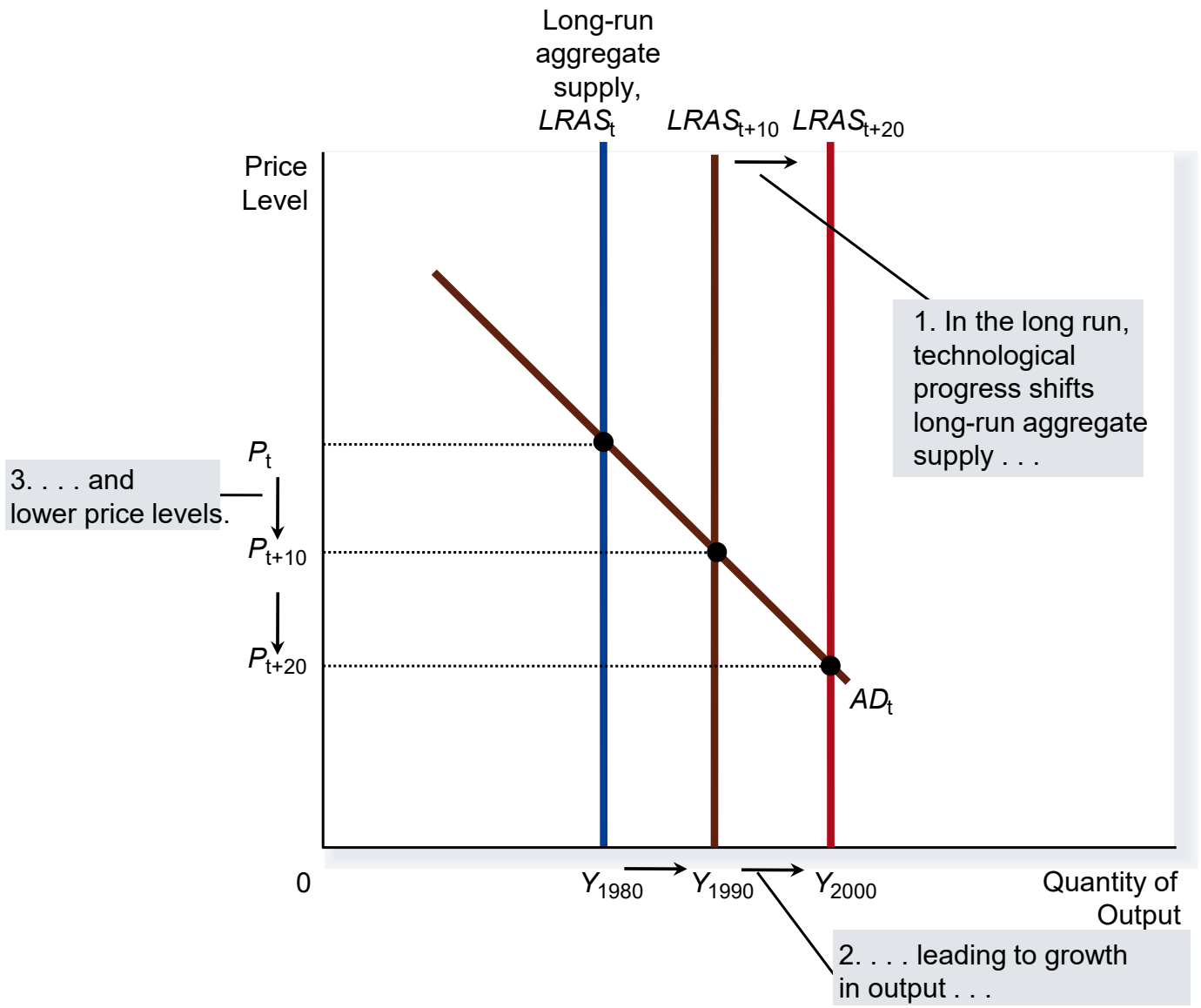
The Long-Run Aggregate-Supply Curve



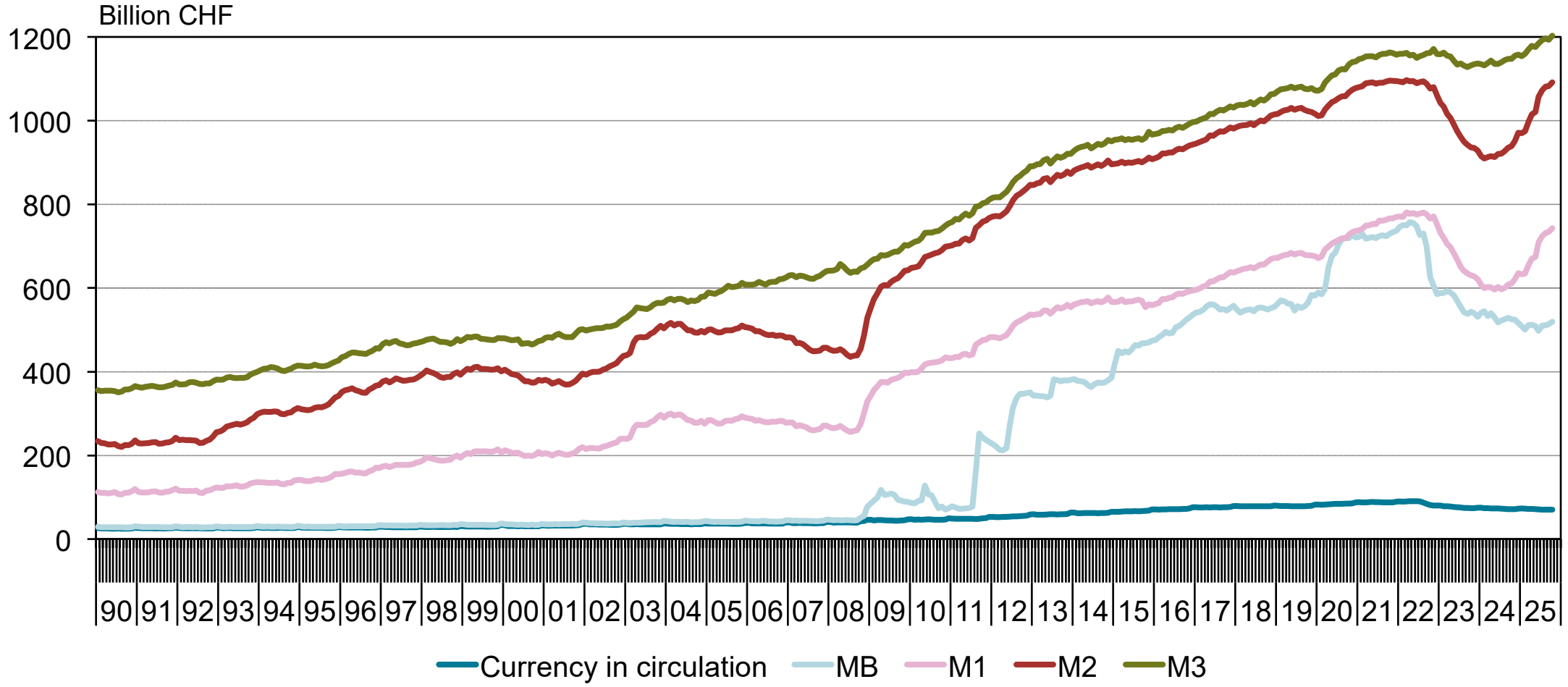
Why the Long-Run Aggregate-Supply Curve Might Shift

- Any change in the economy that alters the natural rate of output shifts the long-run aggregate-supply curve
- The shifts may be categorized according to the various factors in the classical model that affect output
- Shifts might arise from changes in:
  - Labour (e.g. net immigration)
  - Capital
  - Natural Resources
  - Technological Knowledge (i.e. inventions of new technology)

Long-Run Growth and Inflation

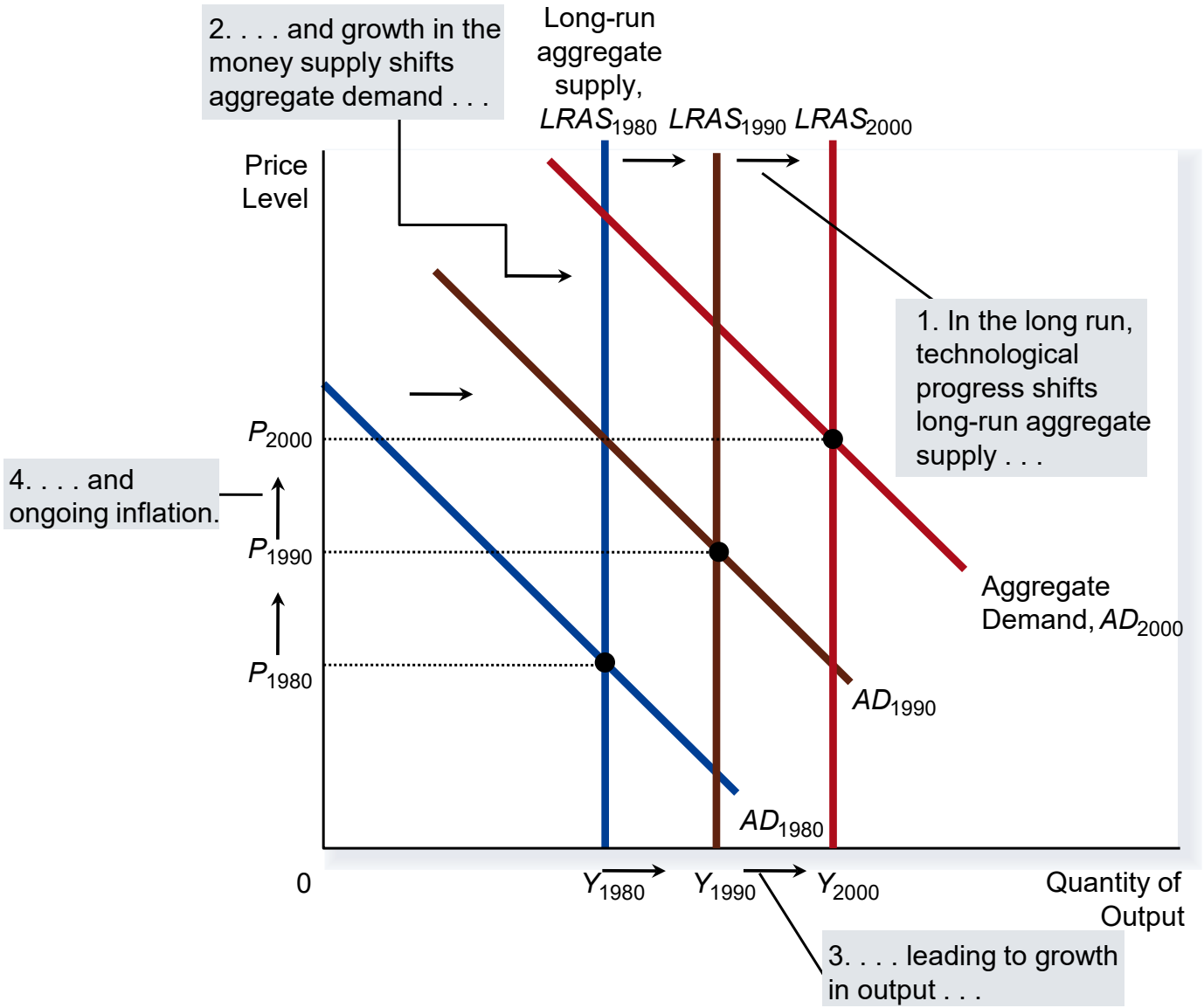


Remember: Money supply in Switzerland



Source: SNB

Long-Run Growth and Inflation



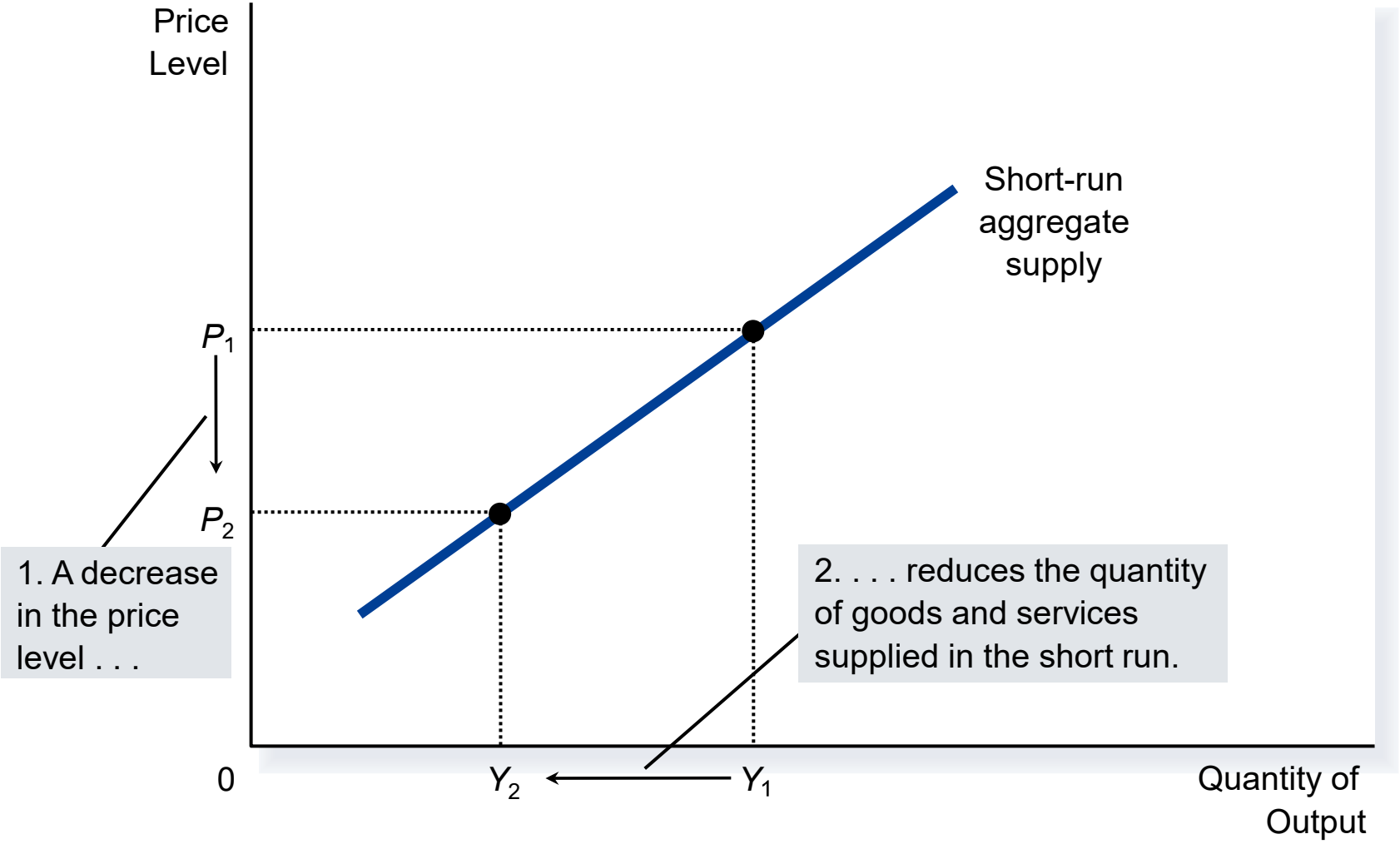
Using Aggregate Demand and Aggregate Supply to Depict Long-Run Growth and Inflation

- The most important forces that govern the economy in the long run are technology (output) and monetary policy (price level)
- Short-run fluctuations in output and the price level should be viewed as deviations from the continuing long-run trends of output growth and inflation

Why the Aggregate-Supply Curve Slopes Upward in the Short Run

- In the short run, an increase in the overall level of prices in the economy tends to raise the quantity of goods and services supplied
- A decrease in the level of prices tends to reduce the quantity of goods and services supplied
- As a result, the short-run aggregate-supply curve is upward sloping

The Short-Run Aggregate-Supply Curve



Three Theories to Explain Why the Aggregate-Supply Curve Slopes Upward in the Short Run

1. The Sticky-Wage Theory
   - Nominal wages are slow to adjust to changing economic conditions, or are “sticky” in the short run
   - A lower price level makes employment and production less profitable
   - This induces firms to reduce the quantity of goods and services supplied
2. The Sticky-Price Theory
   - Prices of some goods and services adjust sluggishly in response to changing economic conditions
   - An unexpected fall in the price level leaves some firms with higher-than-desired prices
     - For a variety of reasons, they may not want to or be able to change prices immediately
   - This depresses (expected) sales, inducing firms to reduce the quantity they produce
3. The Misperceptions Theory
   - Changes in the overall price level temporarily mislead suppliers about what is happening in the markets in which they sell their output
   - A lower price level causes misperceptions about relative prices
   - These misperceptions induce suppliers to decrease the quantity of goods and services supplied

Why the Aggregate-Supply Curve Slopes Upward in the Short Run

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- All these theories suggest that output deviates in the short run from the natural rate when the actual price level deviates from the price level that people had expected to prevail

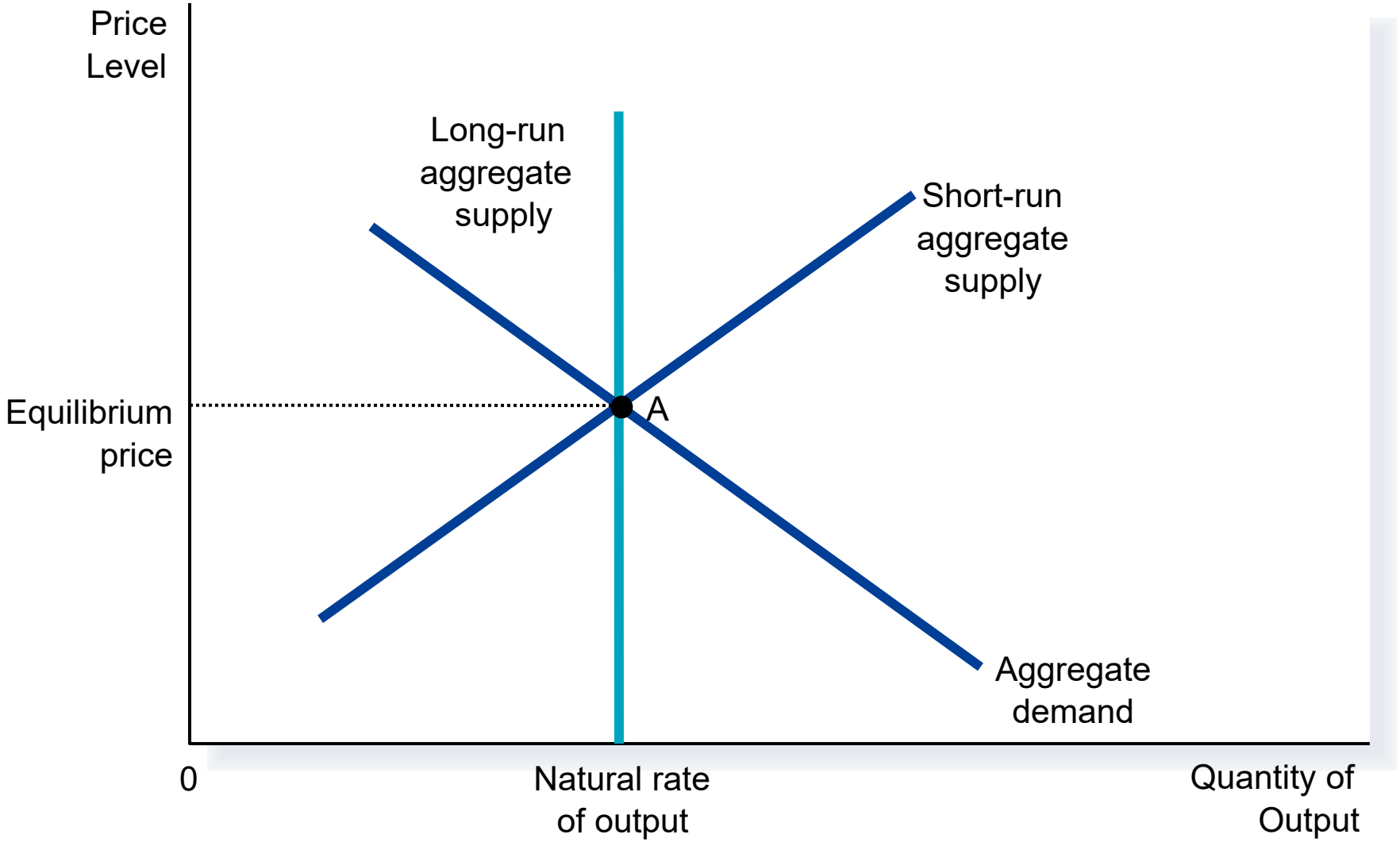
$$\begin{matrix}\textbf{Quantity of}\\ \textbf{Output Supplied}\end{matrix} = \begin{matrix}\textbf{Natural Rate}\\ \textbf{of Output}\end{matrix} + a \times \left(\begin{matrix}\textbf{Actual}\\ \textbf{Price Level}\end{matrix} - \begin{matrix}\textbf{Expected}\\ \textbf{Price Level}\end{matrix}\right)$$

$$\boldsymbol{Y} = \widehat{\boldsymbol{Y}} + a \times (\boldsymbol{P} - \boldsymbol{P^e})$$

Why the Short-Run Aggregate-Supply Curve Might Shift

- Events that shift the long-run aggregate supply curve will shift the short-run aggregate supply curve as well
  - Labour
  - Capital
  - Natural Resources
  - Technology
- People's ***expectations*** of the price level will affect the position of the short-run aggregate supply curve even though it has no effect on the long-run aggregate supply curve.
  - An increase in the expected price level reduces the quantity of goods and services supplied and shifts the short-run aggregate supply curve to the left
  - A decrease in the expected price level raises the quantity of goods and services supplied and shifts the short-run aggregate supply curve to the right

The Long-Run Equilibrium



Summary

- All societies experience short-run economic fluctuations around long-run trends
  - These fluctuations are irregular and largely unpredictable
- When recessions occur, real GDP and other measures of income, spending, and production fall, and unemployment rises
- Economists analyse short-run economic fluctuations using the aggregate demand and aggregate supply model
  - According to the model of aggregate demand and aggregate supply, the output of goods and services and the overall level of prices adjust to balance aggregate demand and aggregate supply
- The aggregate demand curve slopes downward for three reasons:
  - A wealth effect, an interest rate effect, and an exchange rate effect
  - Any event or policy that changes consumption, investment, government purchases, or net exports at a given price level will shift the aggregate demand curve
- In the long run, the aggregate supply curve is vertical
- The short-run, the aggregate supply curve is upward sloping

Summary

- Three theories explain the upward slope of short-run aggregate supply:
  - 1. The misperceptions theory,
    2. the sticky wage theory, and
    3. the sticky price theory
  - Events that alter the economy's ability to produce will shift the short-run aggregate supply curve
  - The position of the short-run aggregate supply curve depends on the expected price level
- One possible cause of economic fluctuations is a shift in aggregate demand
- A second possible cause of economic fluctuations is a shift in aggregate supply
- Stagflation is a period of falling output and rising prices

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Principles of Macroeconomics

Lecture 12: Monetary and Fiscal Policy

Prof. Dr. Jan-Egbert Sturm

Analysing economic fluctuations: the AD/AS Model

- Aggregate demand

$$\boldsymbol{Y} = \boldsymbol{C} + \boldsymbol{I} + \boldsymbol{G} + \boldsymbol{NX}$$

$$\boldsymbol{M}/\boldsymbol{P} = L(\boldsymbol{i}, \boldsymbol{Y}) \quad \text{or} \quad \boldsymbol{M} \times \boldsymbol{V}(\boldsymbol{i}) = \boldsymbol{P} \times \boldsymbol{Y}$$

- (Short-run) Aggregate supply

$$\boldsymbol{Y} = \widehat{\boldsymbol{Y}} + a \times (\boldsymbol{P} - \boldsymbol{P}^{\boldsymbol{e}})$$

- Four steps in the process of analysing economic fluctuations:
  1. Determine whether the event affects aggregate supply or aggregate demand
  2. Decide which direction the curve shifts
  3. Use a diagram to compare the initial and the new equilibrium
  4. Keep track of the short and long run equilibrium, and the transition between them

Two Causes of Economic Fluctuations

1. Shifts in Aggregate Demand
2. Shifts in Aggregate Supply

Two Causes of Economic Fluctuations

1. Shifts in Aggregate Demand
   - In the short run, shifts in aggregate demand cause fluctuations in the economy's output of goods and services
   - In the long run, shifts in aggregate demand affect the overall price level but do not affect output
   - Policymakers who influence aggregate demand can potentially mitigate the severity of economic fluctuations
2. Shifts in Aggregate Supply

The Influence of Monetary and Fiscal Policy on Aggregate Demand

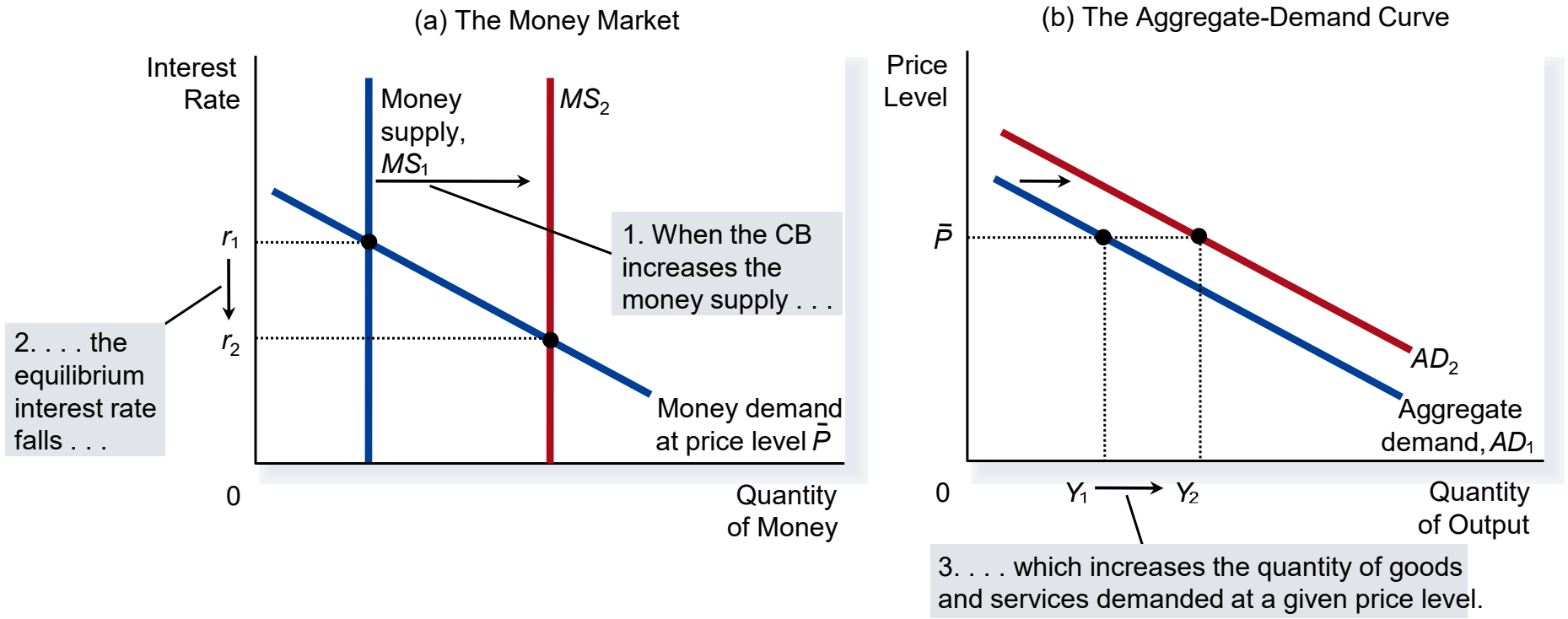
- Many factors influence aggregate demand besides monetary and fiscal policy
- In particular, desired spending by households and business firms determines the overall demand for goods and services
- When desired spending changes, aggregate demand shifts, causing short-run fluctuations in output and employment
- ***Monetary*** and ***fiscal*** policy are sometimes used to offset those shifts and stabilize the economy

Monetary Policy: Changes in the Money Supply

- The Central Bank can shift the aggregate demand curve when it changes ***monetary policy***
- An increase in the money supply shifts the money supply curve to the right
- Without a change in the position of the money demand curve, the interest rate falls
- Falling interest rates increase the quantity of goods and services demanded

- When the Central Bank increases the money supply, it lowers the interest rate and increases the quantity of goods and services demanded at any given price level, shifting aggregate-demand to the right
- When the Central Bank decreases the money supply, it raises the interest rate and reduces the quantity of goods and services demanded at any given price level, shifting aggregate-demand to the left

Monetary Policy: A Monetary Injection



The Role of Interest-Rate Targets in Central Bank Policy

- Monetary policy can be described either in terms of the money supply or in terms of the interest rate
- Changes in monetary policy can be viewed either in terms of a changing target for the interest rate or in terms of a change in the money supply
- A target for the federal funds rate affects the money market equilibrium, which influences aggregate demand

How Fiscal Policy Influences Aggregate Demand

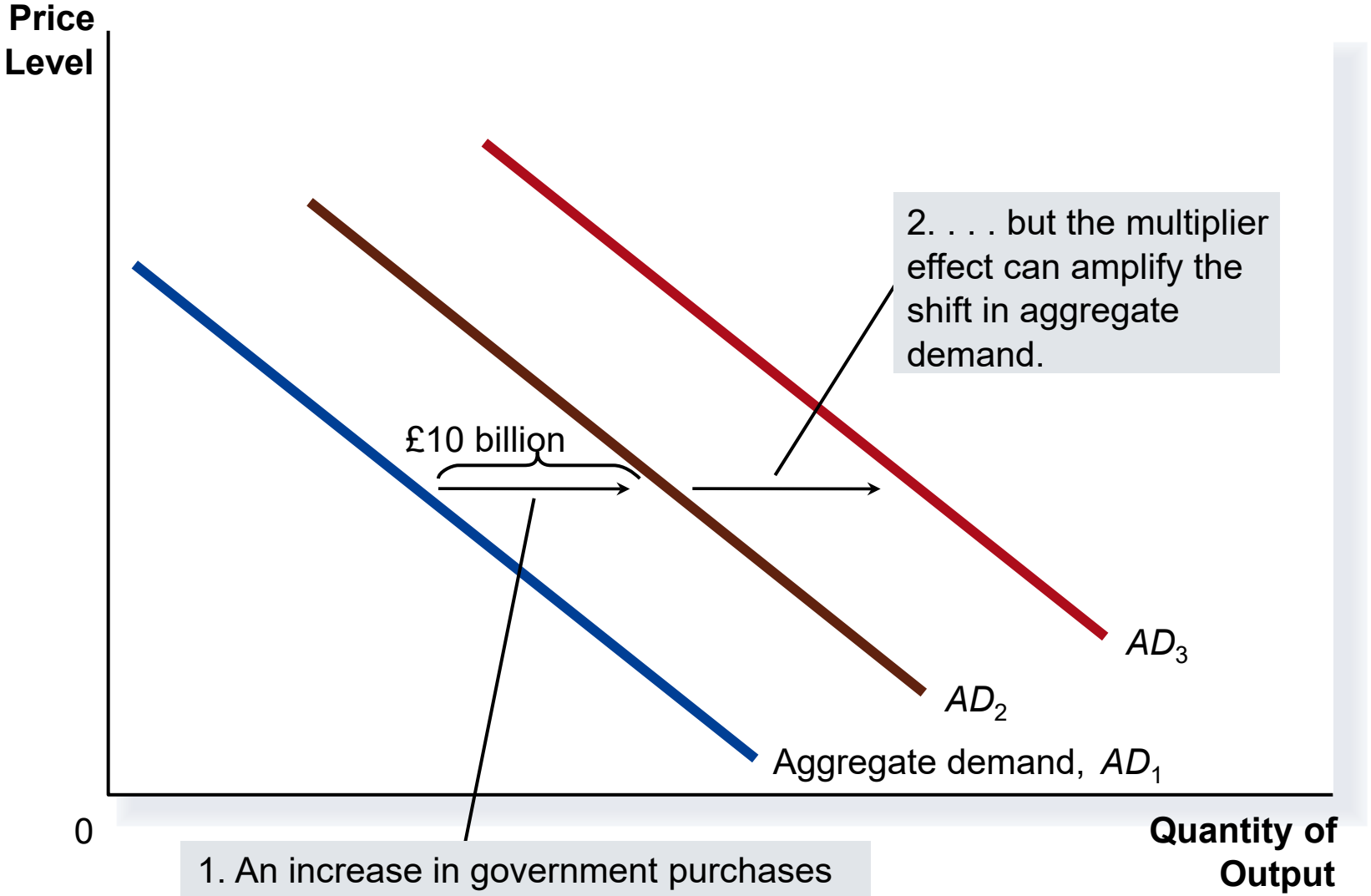
- ***Fiscal policy*** refers to the government's choices regarding the overall level of government purchases or taxes
- Fiscal policy influences saving, investment, and growth in the long run
- In the short run, fiscal policy primarily affects the aggregate demand

- When policymakers change the money supply (M) or taxes (T),
  the effect on aggregate demand is indirect—through the spending decisions of firms or households
- When the government alters its own purchases of goods or services (G),
  it shifts the aggregate-demand curve directly (as it directly part of the national income account identity)
- There are two (3) macroeconomic effects from a change in government purchases (or taxes):
  1. The multiplier effect
  2. The crowding-out effect
  3. (Ricardian equivalence: People anticipate future tax changes and change own savings accordingly)

Changes in Government Purchases: Multiplier Effect

- Government purchases are said to have a ***multiplier effect*** on aggregate demand
  - Each pound spent by the government can raise the aggregate demand for goods and services by more than a pound
  - The multiplier effect refers to the additional shifts in aggregate demand that result when expansionary fiscal policy increases income and thereby increases consumer spending
    - $Multiplier = 1/(MPS + MPT + MPM) = 1/MPW \qquad (W = \text{Withdrawal})$

Changes in Government Purchases: Multiplier Effect

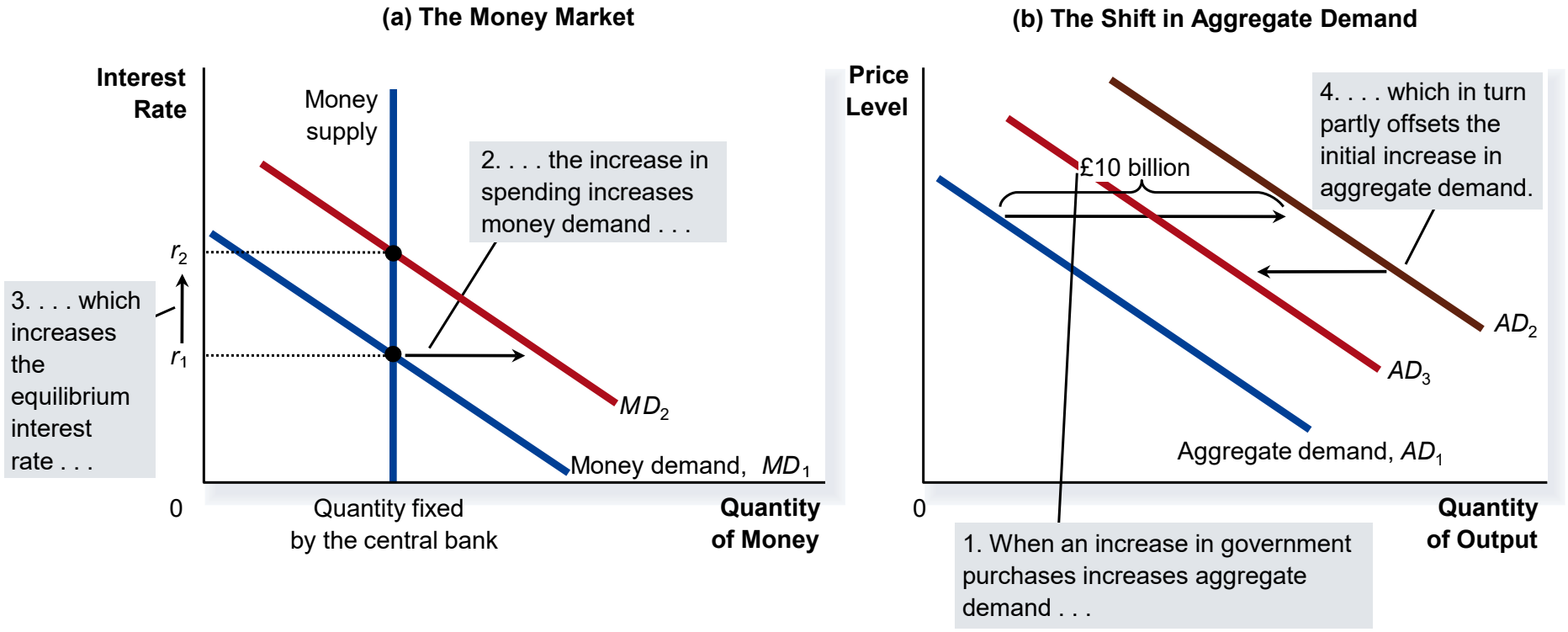


1. An increase in government purchases of £10 billion initially increases aggregate demand by £10 billion . . .

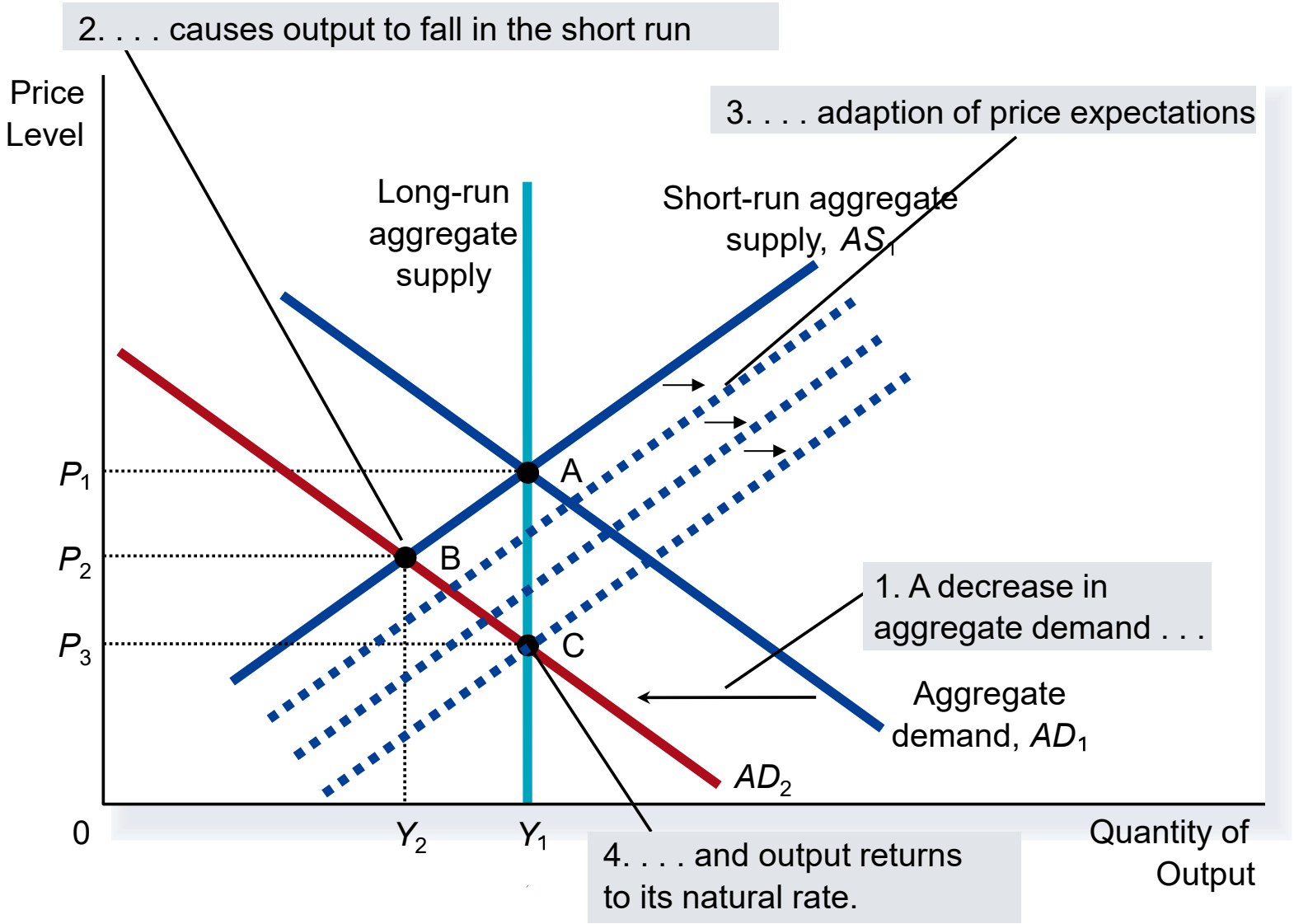
Changes in Government Purchases: Crowding-Out Effect

- Fiscal policy may not affect the economy as strongly as predicted by the multiplier
  - An increase in government purchases causes the interest rate to rise
  - A higher interest rate reduces investment spending
- This reduction in demand that results when a fiscal expansion raises the interest rate is called the ***crowding-out effect***
  - The crowding-out effect tends to dampen the effects of fiscal policy on aggregate demand
- When the government increases its purchases by £10 billion,
  the aggregate demand for goods and services could rise by more or less than £10 billion,
  depending on whether the multiplier effect or the crowding-out effect is larger

Changes in Government Purchases: Crowding-Out Effect



Changes in Taxes

- When the government cuts personal income taxes, it increases households' take-home pay
- Households save some of this additional income
- Households also spend some of it on consumer goods
- Increased household spending shifts the aggregate-demand curve to the right
- The size of the shift in aggregate demand resulting from a tax change is affected by the multiplier and crowding-out effects
- It is also determined by the households' perceptions about the permanency of the tax change

A Contraction in Aggregate Demand with Policy Reacting